Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
SUBJECT CODE:	**SECR 1033**
SEMESTER:	**2 – 2023/2024**
LAB TITLE:	**Lab 2: Arithmetic Equations & Operations**
STUDENT INFO :	Execute the lab in group of two. **Student 1**: *No. 1, 3, 5* **Student 2**: *No 2, 4, 6* **Name 1:** **Koo Xuan** **Metric No:** **A23CS0300** **Link for Video Demo:** **https://drive.google.com/drive/folders/1J2wIdsrMmOF8mJ94kWV3s7L5tYR1zMHl?usp=sharing** **Name 2:** **Ling Yu Qian** **Metric No:** **A23CS0301** **Link for Video Demo:** **https://drive.google.com/drive/folders/1J2wIdsrMmOF8mJ94kWV3s7L5tYR1zMHl?usp=sharing**
SUBMISSION DATE & ITEMS:	**Duration of submission :-** 2 weeks **Submission items in elearning:-**

	1. Lab 2 exercise sheet/file (in .pdf), with the links for demo video 5 - 10min, on the cover page. 2. The assembly programs (in .asm).

MARKS:
_____

Arithmetic Equation Coding in Assembly Language

Q1. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 1 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

```
INCLUDE Irvine32.inc
.data
var1 word 1
var2 word 9

.code
main PROC
        mov ax, var1     ; LINE1
        mov bx, var2    ; LINE2
        xchg ax, bx      ; LINE3
        mov var1, ax     ; LINE4
        mov var2, bx     ; LINE5
        call DumpRegs
        exit
main ENDP
END main
```

Answer Q1

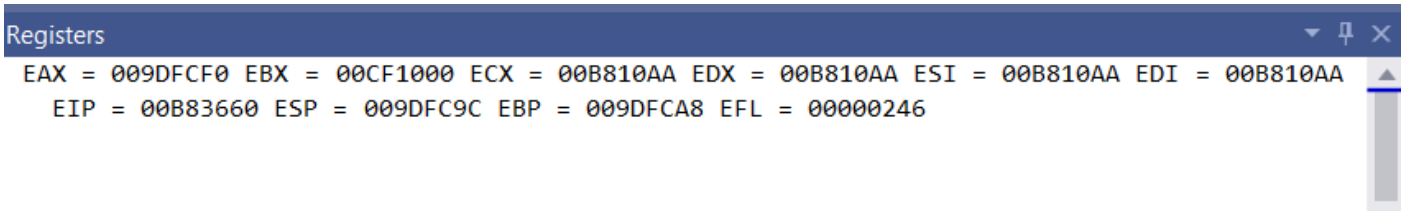
a) Fill (Write) in the contents for the related register in each line:

Table 1

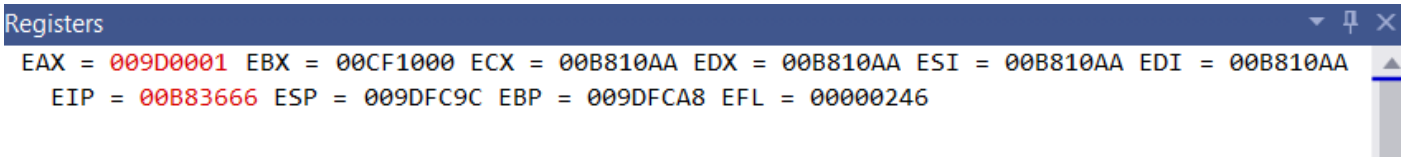
LINE1	**AX = 0001h** **var1 = 0001h**	Move the value of var1 (1d) into register AX
LINE2	**BX** = 0009h **var2** = 0009h	Move the value of var2 (9d) into register BX
LINE3	**AX** = 0009h **BX** = 0001h	AX ↔ BX exchange 16-bit registers
LINE4	**AX** = 0009h **var1** = 0009h	Move the value of register AX after exchange which is (0009h) into var1
LINE5	**BX** = 0001h **var2** = 0001h	Move the value of register BX after exchange which is (0001h) into var2

b) Paste here screenshot of all registers' content after each LINE is executed:

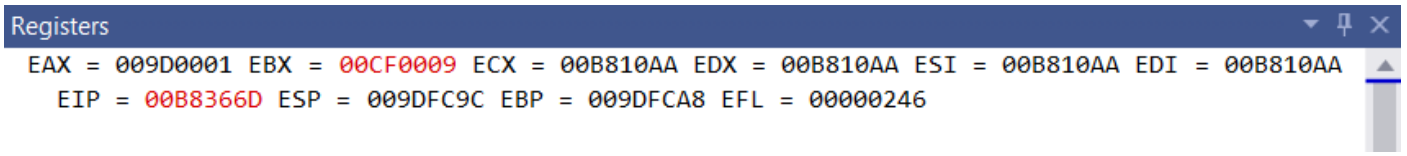
LINE1:



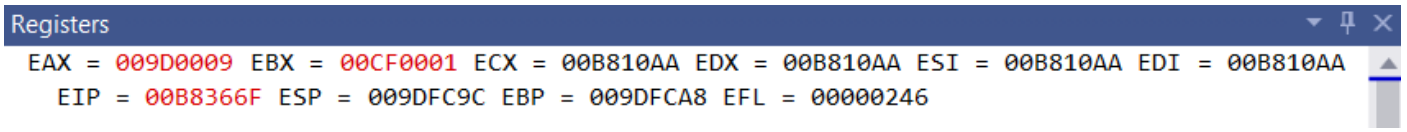
LINE2:



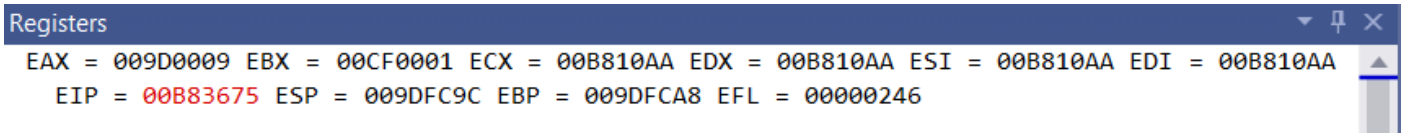
LINE3:



LINE4:



LINE5:



Q2. Execute the program below. Determine output of the program by inspecting the content of the related registers and watches.

a) Fill in Table 2 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: Rval = (-Xval + (Yval – Zval)) + 1

```
include irvine32.inc

.data
Rval DWORD ?
Xval DWORD 26
Yval DWORD 30
Zval DWORD 40

.code
main proc
      mov eax,Xval ; LINE1
      neg eax            ; LINE2
      mov ebx,Yval ; LINE3
      sub ebx,Zval ; LINE4
      add eax,ebx        ; LINE5
      inc eax            ; LINE6
      mov Rval,eax ; LINE7
      exit
main endp
end main
```

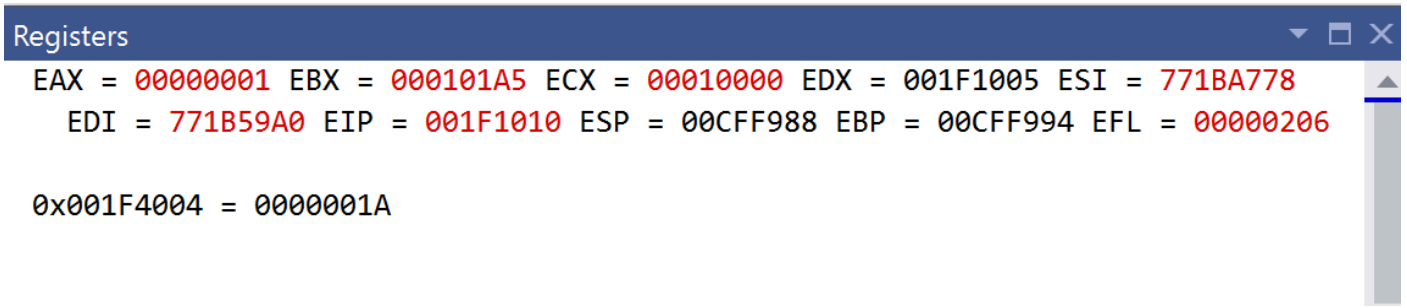
Answer Q2

a) Fill (Write) in the contents for the related register in each line:

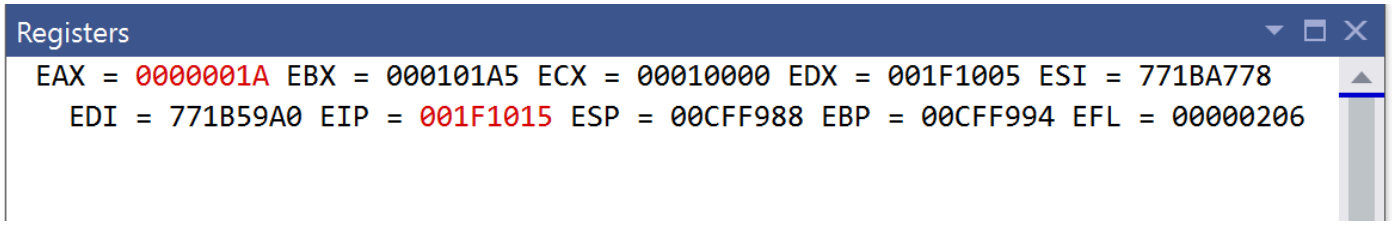
LINE1	EAX = 0000001Ah Xval = 0000001Ah	Move the value of Xval (26d) into register EAX
LINE2	EAX = FFFFFFE6h	Negate register EAX with two's complement
LINE3	EBX = 0000001Eh Yval = 0000001Eh	Move the value of Yval (30d) into register EBX
LINE4	EBX = FFFFFFF6h Zval = 00000028h	Subtract value of Zval (40d) from register EBX
LINE5	EAX = FFFF FFDCh EBX = FFFF FFF6h	Add value of register EBX to register EAX
LINE6	EAX = FFFF FFDDh	Increment of EAX by 1
LINE7	EAX = FFFF FFDDh Rval = FFFF FFDDh	Move value store in register EAX into variable Rval.

b) Paste here screenshot of all registers' content after each LINE is executed:

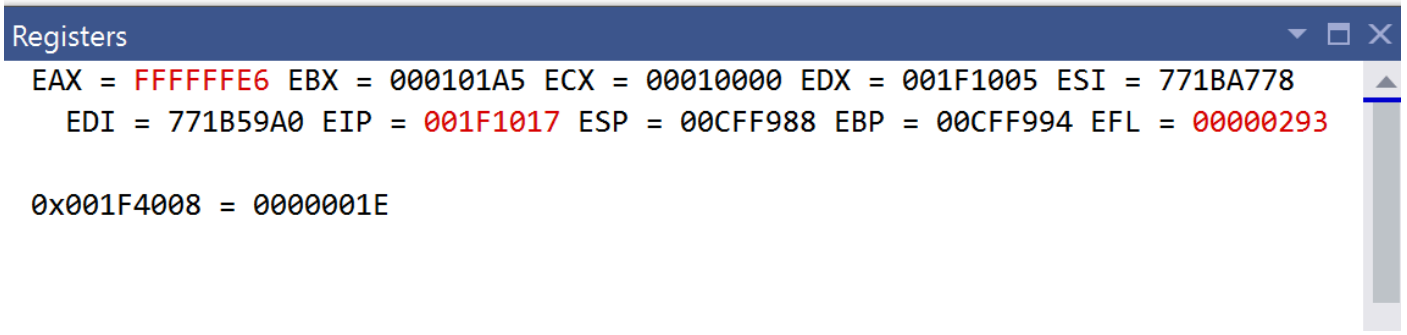
LINE1:



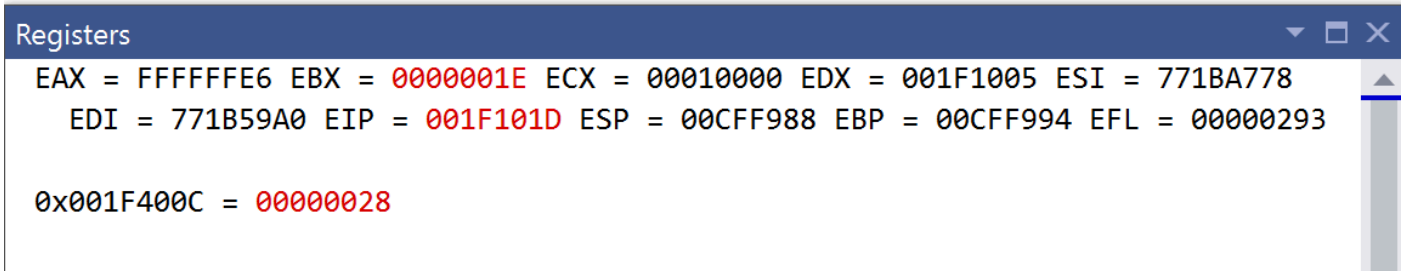
LINE2:



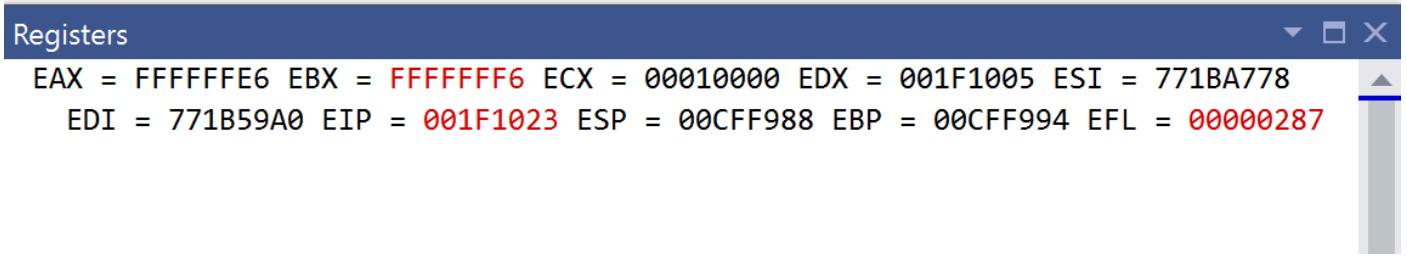
LINE3:



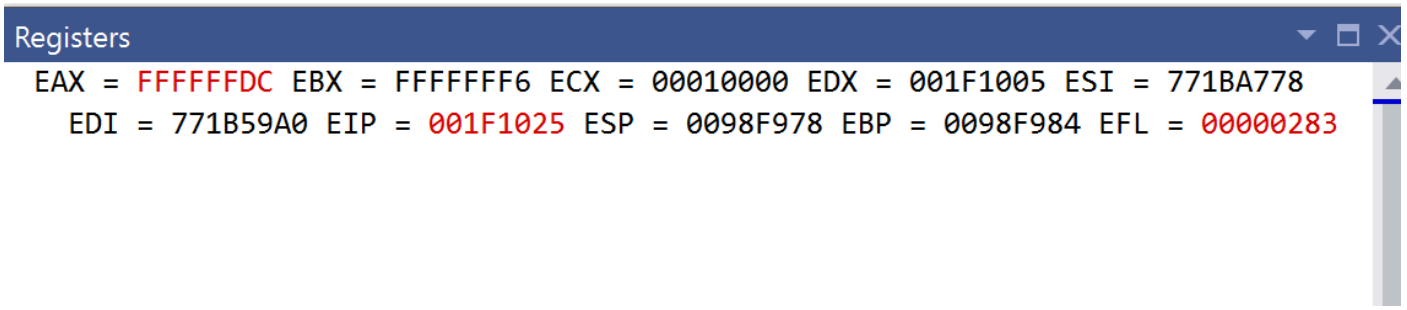
LINE4:



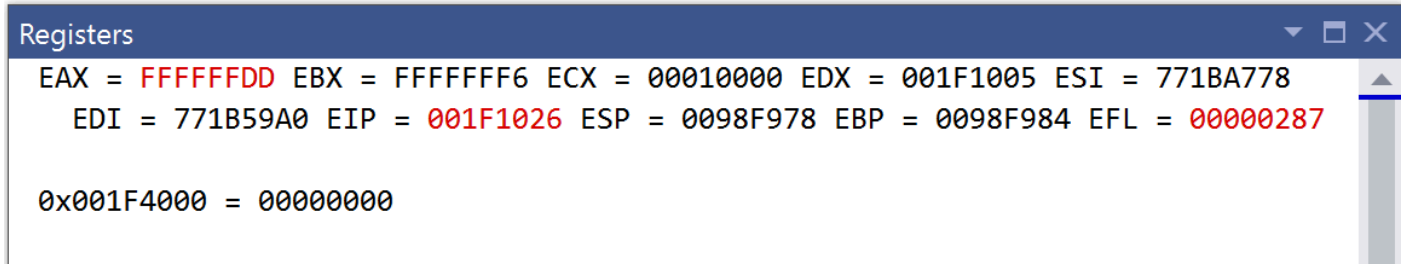
LINE5:



LINE6:



LINE7:



Q3. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 3 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = [(var1 * var2) + var3] - 1

```
include irvine32.inc

.data
var1 DWORD 5
var2 DWORD 10
var3 DWORD 20
var4 DWORD ?

.code
main proc
      mov eax, var1          ; LINE1
      mul var2               ; LINE2
      add eax, var3          ; LINE3
      dec eax                ; LINE4
      exit
main endp
end main
```

Answer Q3

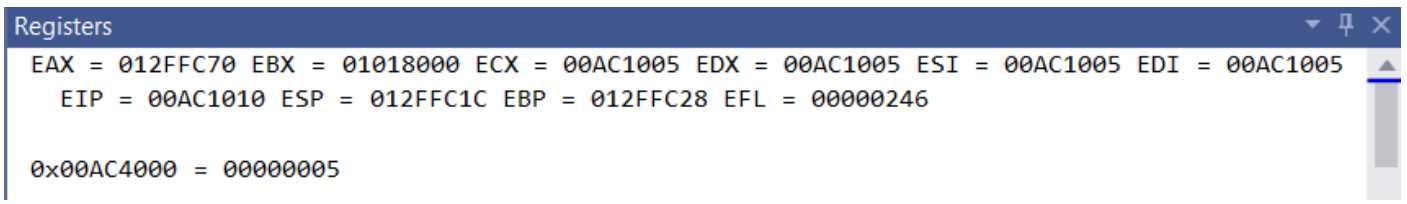
a) Fill (Write) in the contents for the related register in each line:

Table 3

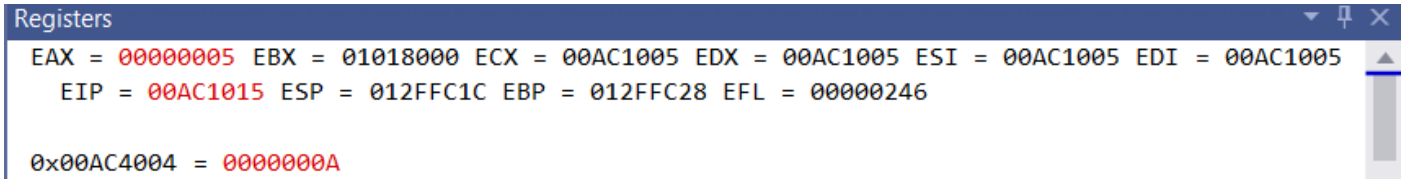
LINE1	**EAX = 00000005h** **var1 = 00000005h**	Move the value of var1 (5d) into register EAX
LINE2	**EAX =** 00000032h **var2 =** 0000000Ah	Multiply the value of var2 (10d) into register EAX
LINE3	**EAX =** 00000046h **var3 =** 00000014h	Add the value of var3 (20d) to register EAX
LINE4	**EAX =** 00000045h **var4 =** 00000045h	Subtract 1 from the single operand. Move the value of register EAX into variable var4

b) Paste here screenshot of all registers' content after each LINE is executed:

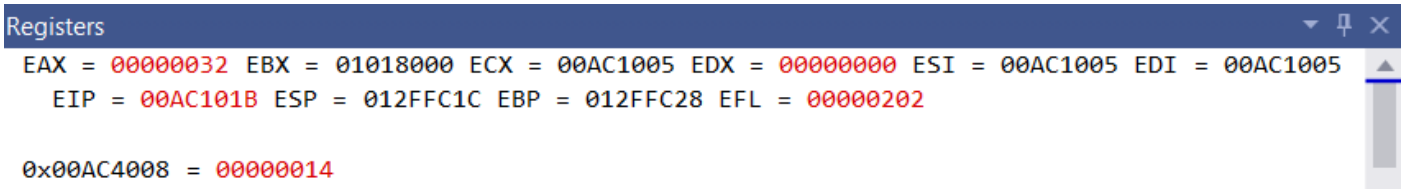
LINE1:



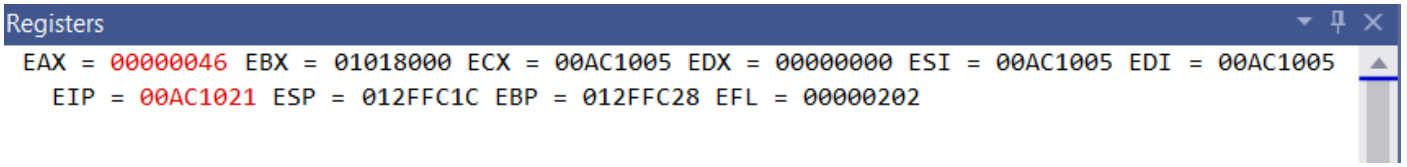
LINE2:



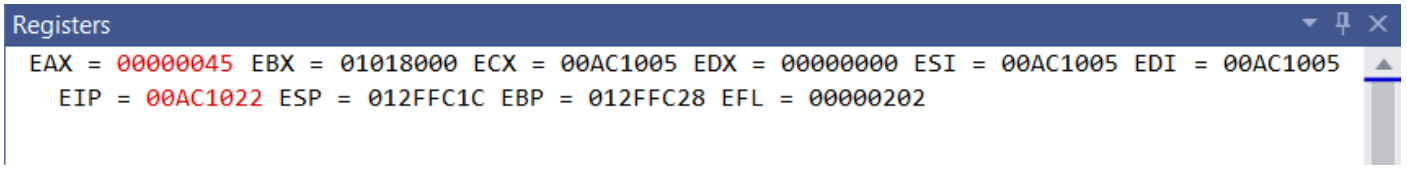
LINE3:



LINE4:



After LINE 4:



Q4. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 4 with the content of each register or variable on every LINE, in Hexadecimal (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = (var1 * 5) / (var2 – 3)

```
include irvine32.inc
.data
      var1 WORD 40
      var2 WORD 10
      var4 WORD ?
.code
main proc
      mov ax,var1             ; LINE1
      mov bx,5          ; LINE2
      mul bx            ; LINE3
      mov bx,var2             ; LINE4
      sub bx,3          ; LINE5
      div bx            ; LINE6
      mov var4,ax             ; LINE7
      exit
main endp
end main
```

Answer Q4

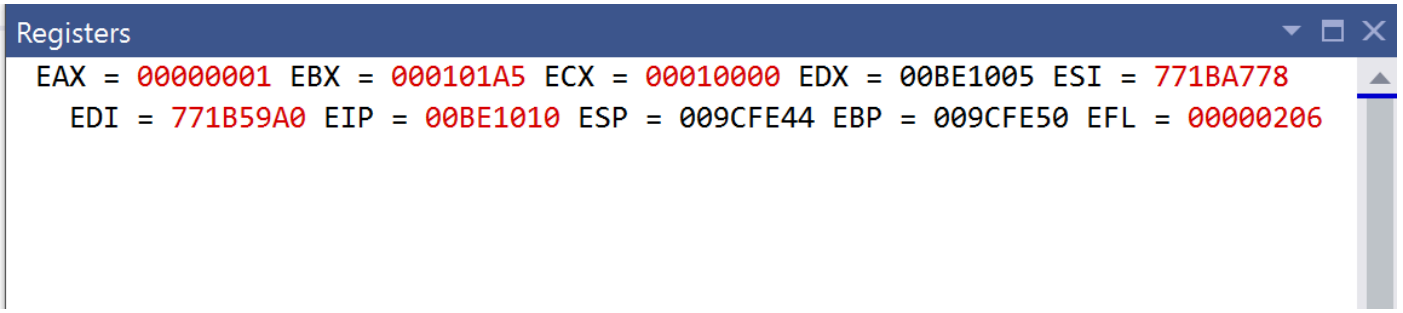
a) Fill (Write) in the contents for the related register in each line:

Table 4

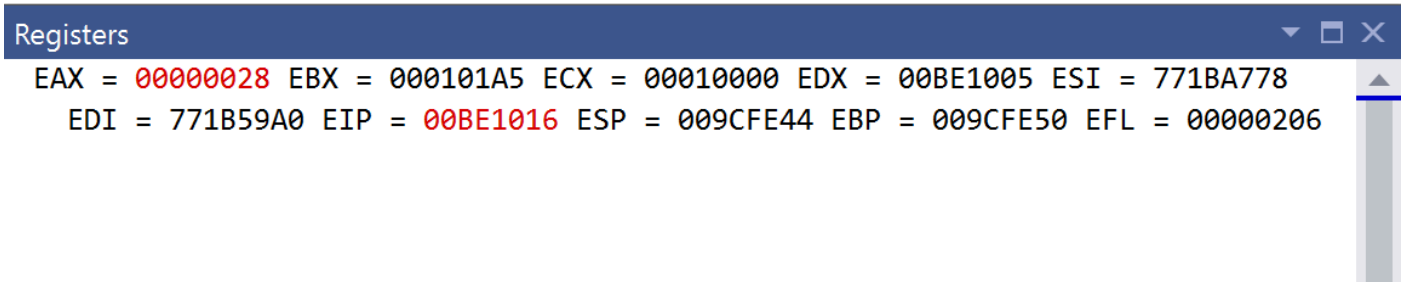
LINE1	**AX = 0028h** **var1 = 0028h**	Move the value of var1 (40d) into register AX
LINE2	**BX =** 0005h	Move the immediate value(5d) into register BX
LINE3	**AX =** 00C8h **BX =** 0005h	Multiply the value of register BX to register AX and store the value in register DX:AX
LINE4	**BX =** 000Ah **var2 =** 000Ah	Move the value of var2 (10d) into register BX
LINE5	**BX =** 0007h	Subtract the value in register BX with an immediate value (3d)
LINE6	**AX =** 001Ch **BX =** 0007h **DX =** 0004h	Divide the register AX by register BX, the quotient is then store in register AX and the remainder store in register DX.
LINE7	**AX =** 001Ch **var4 =** 001Ch	Move the value of register AX into variable var4.

b) Paste here screenshot of all registers' content after each LINE is executed:

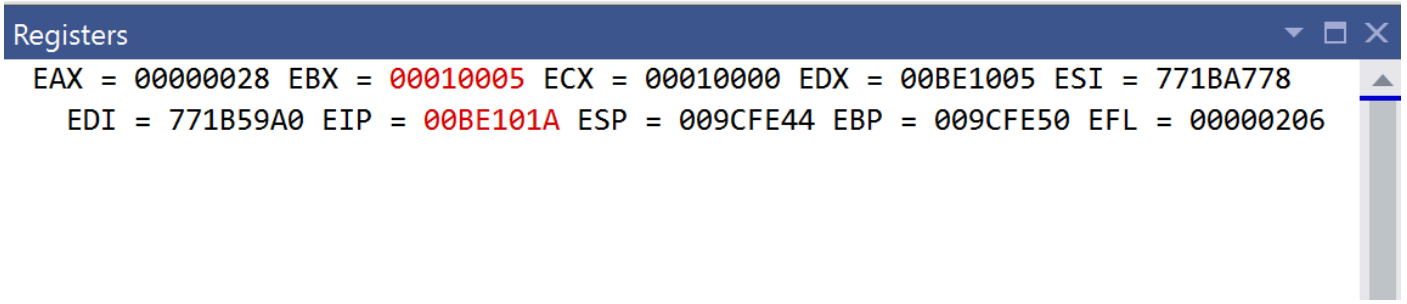
LINE1:



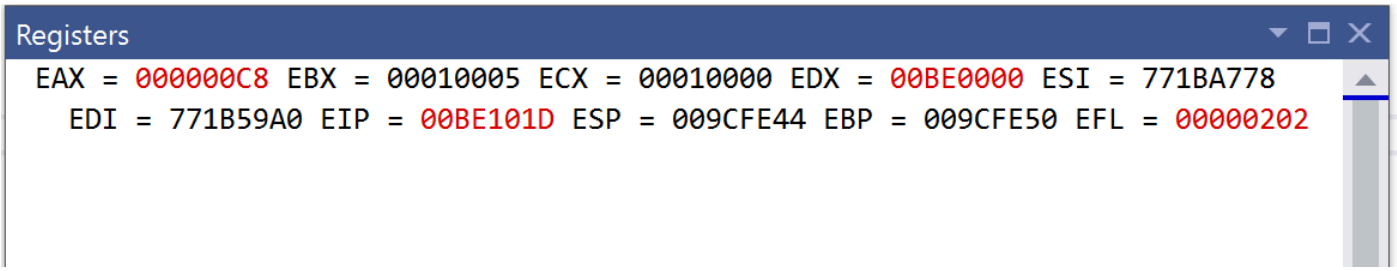
LINE2:



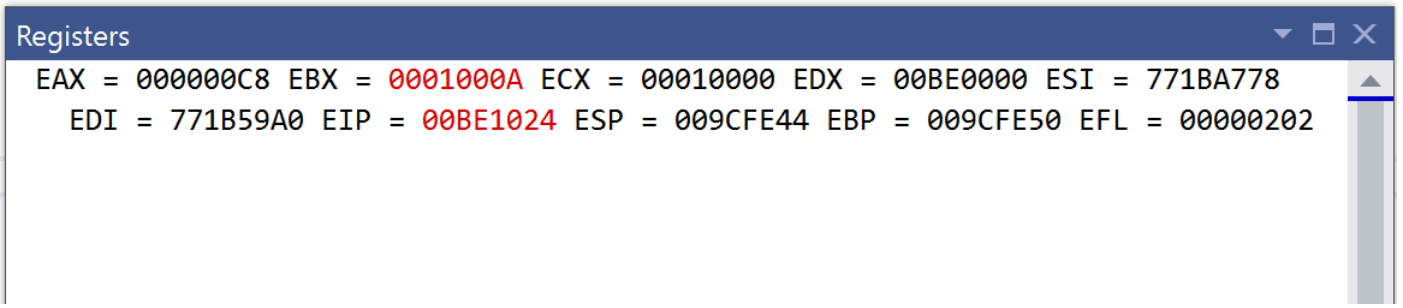
LINE3:



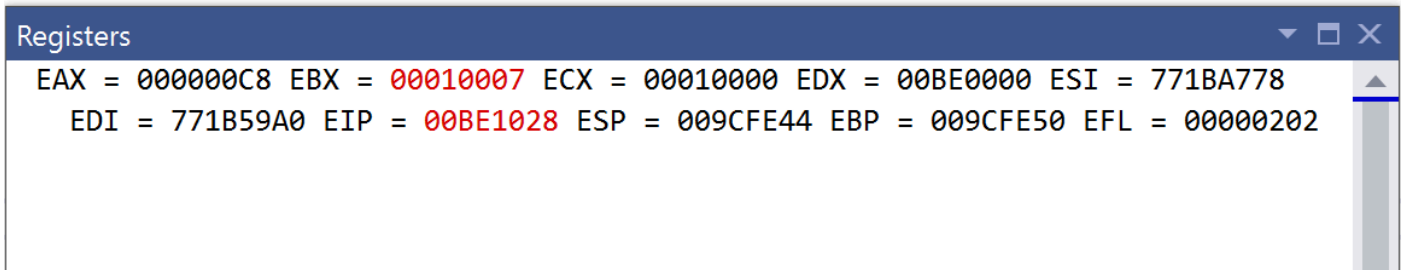
LINE4:



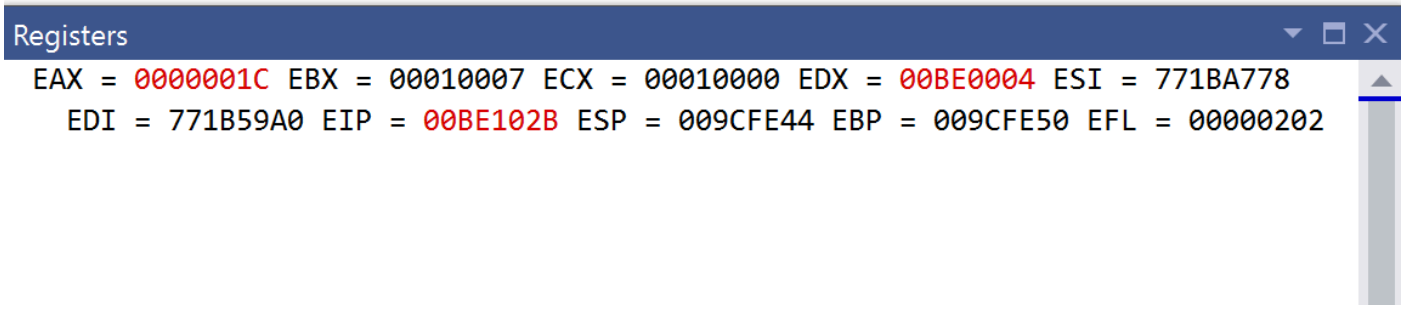
LINE5:



LINE6:



LINE7:



Short Notes for MUL CX and DIV BL:

MUL CX

a. MUL always uses AX (or its extended versions EAX or RAX) as the implicit destination register.

b. The operand size determines the size of the result:

   i. Byte-sized operand: Result in AX

   ii. Word-sized operand: Result in DX:AX

   iii. Doubleword-sized operand (32-bit mode): Result in EDX:EAX

   iv. Quadword-sized operand (64-bit mode): Result in RDX:RAX

c. The upper half of the result (DX or EDX or RDX) holds any overflow bits.

d. The Carry Flag (CF) is set if the upper half of the product is non-zero.

DIV BL

a. DIV always uses the DX:AX or EDX:EAX pair as the implicit dividend register.

b. The divisor is specified as the operand of the DIV instruction.

c. The quotient is stored in AX (for 16-bit division) or EAX (for 32-bit division).

d. The remainder is stored in DX.

e. Clear DX (or EDX for 32-bit division) before division to ensure a correct 16-bit or 32-bit dividend.

f. If the divisor is 0, a division error occurs.

g. The Overflow Flag (OF) is set if the quotient is too large to fit in the destination register.

Q5. Given the following instructions as is Code Snippet 1.

a) Write a full program to execute the Code Snippet 1.
b) What are the contents of the related registers after Code Snippet 1 is executed? Paste the screenshot of DumpReg.

```
; Code Snippet 1 (MUL CX)

MOV DX,0        ; Clear DX
MOV AX,1000h    ; Load 1000h into AX
MOV CX, 25h     ; Load 25h into CX
MUL CX          ; Multiply AX by CX, storing the result in DX:AX
```

Answer Q5

a) Screenshot of full program (.asm) :

main.asm

lab2_Question5 (Global Scope)

```
1   TITLE lab2_Question5
2   ; Author: Koo Xuan& Ling Yu Qian
3   ; Date: 7 JUNE 2024
4
5   include Irvine32.inc
6
7   .data
8   ;no data in this program
9
10  .code
11  main proc
12  MOV DX, 0       ; Clear DX
13  MOV AX, 1000h   ; Load 1000h into AX
14  MOV CX, 25h     ; Load 25h into CX
15  MUL CX          ; Multiply AX by CX, storing the result in DX : AX
16  call DumpRegs
17  exit
18
19  main ENDP
20
21  END main
```

b) Paste here the screenshot of the final registers' content (DumpReg):

Microsoft Visual Studio Debu

```
EAX=00575000  EBX=003BD000  ECX=00430025  EDX=00430002
ESI=004310AA  EDI=004310AA  EBP=0057FF2C  ESP=0057FF20
EIP=00433674  EFL=00000A07  CF=1  SF=0  ZF=0  OF=1  AF=0  PF=1

C:\Users\user\OneDrive\Documents\Visual Studio 2019\Lab work\Lab2_Question5\lab2_Question5\Debug\lab2_Question5.exe (pro
cess 8532) exited with code 0.
Press any key to close this window . . .
```

Q6. Given the following instructions as is Code Snippet 2.

a) Write a full program to execute the Code Snippet 2.
b) What are the contents of the related registers after Code Snippet 2 is executed? Paste the screenshot of DumpReg.

```
; Code Snippet 2 (DIV BL)
MOV DX, 0       ; Clear DX to form the 16-bit dividend in DX:AX
MOV AX, 803h    ; Load the dividend (8003h) into AX
MOV BL, 10h     ; Load the divisor (10h) into BL
DIV BL          ; Divide DX:AX by BL, whereby AX=quotient & DX=remainder
```

Answer Q6

Screenshot of full program (.asm) :

LAB2Q6 (Global Scope)

```
1  ; Title: lab2
2  ; Author: koo xuanand ling yu qian
3  ; Date: 7 / 6 / 2024
4
5  include irvine32.inc
6
7  .data
8  ; No data section needed for this example
9
10 .code
11 main proc
12 ; Clear DX to form the 16 - bit dividend in DX : AX
13 mov dx, 0
14
15 ; Load the dividend(0803h) into AX
16 mov ax, 0803h
17
18 ; Load the divisor(10h) into BL
19 mov bl, 10h
20
21 ; Divide DX : AX by BL, resulting in AX = quotient, DX = remainder
22 div bl
23
24 ; Dump register contents to see the result
25 call DumpRegs
26
27 ; Exit the program
28 exit
29
30 main endp
31 end main
32
```

Paste here the screenshot of the final registers' content (DumpReg):

Microsoft Visual Studio Debu

```
EAX=00000380  EBX=00010110  ECX=00010000  EDX=00DE0000
ESI=771BA778  EDI=771B59A0  EBP=00AFF980  ESP=00AFF974
EIP=00DE3671  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1

C:\Users\User\source\repos\LAB2Q6\Debug\LAB2Q6.exe (process 26556) exited with code 0.
Press any key to close this window . . .
```